Agri Guardian: An Al-Powered Plant Disease Diagnosis with Decision Support System

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Abstract*—The use of artificial intelligence (AI) in the detection and diagnosis of plant diseases has gained significant interest in modern agriculture. The appeal of AI arises from its ability to rapidly and precisely analyze extensive and complex information, allowing farmers and agricultural experts to quickly identify plant diseases. The use of artificial intelligence (AI) in the detection and diagnosis of plant diseases has gained significant attention in the world of agriculture and agronomy. By harnessing the power of AI to identify and diagnose plant diseases, it is expected that farmers and agricultural experts will have improved capabilities to tackle the challenges posed by these diseases. This will lead to increased effectiveness and efficiency, ultimately resulting in higher agricultural productivity and reduced losses caused by plant diseases. The use of artificial intelligence (AI) in the detection and diagnosis of plant diseases has resulted in significant benefits in the field of agriculture. By using AI technology, farmers and agricultural professionals can quickly and accurately identify illnesses affecting their crops. This allows for the prompt adoption of appropriate preventative and corrective actions, therefore reducing losses caused by plant diseases

***Keywords*—Artificial intelligence, plant care, computer vision, plant disease diagnosis, severity estimation, MobileViT, EfficientNet, image segmentation,retrieval augmented generation.**

1.INTRODUCTION

Home gardening has become increasingly accessible through smartphones, online resources, and low-cost growing supplies. However, the availability of information does not guarantee that a beginner can convert that information into correct day-to-day action. A first-time grower may not know which plant is appropriate for available sunlight and space, how to interpret leaf symptoms, or whether a problem is related to water, light, pests, disease, or general stress. This creates a gap between the large amount of agricultural information available online and the practical guidance needed by an individual user.

Deep learning has already demonstrated the feasibility of image-based plant disease recognition. Mohanty, Hughes, and Salathé trained a convolutional neural network using 54,306 PlantVillage images and reported strong performance for classifying crop-disease categories under their experimental setting [1]. However, image classification alone does not constitute a complete plant-care assistant. Agri Guardian therefore treats diagnosis as one component within a larger personalized workflow.

The proposed system begins with user and environmental information, recommends suitable plants, provides setup and care guidance, accepts plant images for identification or health analysis, estimates visible severity, and

retrieves evidence from a curated knowledge base for prevention and treatment guidance. The application is designed for beginners and therefore emphasizes understandable explanations rather than raw model outputs.

II. RELATED WORK

Research on smartphone-assisted plant diagnosis demonstrates that computer vision can reduce the difficulty of identifying visible plant diseases. Mohanty et al. showed the feasibility of deep convolutional networks for crop-disease classification [1]. This work provides a foundation for the diagnosis component of Agri Guardian, but its primary task is classification rather than end-to-end personalized plant care.

EfficientNet introduced a compound scaling strategy that balances network depth, width, and input resolution to improve the accuracy-efficiency trade-off of convolutional models [2]. This makes EfficientNet a candidate backbone for Agri Guardian where model capability and computational cost must both be considered.

MobileViT combines convolutional processing with transformer-based representations and was proposed as a lightweight architecture for mobile vision tasks [3]. Its mobile-oriented design makes it a useful candidate for comparison with conventional CNN backbones during model selection.

For affected-region analysis, segmentation is preferable to relying only on a disease-class label. U-Net introduced an encoder-decoder architecture with a contracting path for context and an expanding path for precise localization [4]. A similar segmentation strategy can be investigated for estimating the visible portion of a leaf affected by a detected condition.

YOLO established a fast unified approach to object detection by predicting localization and class probabilities in a single network [5]. Detection or segmentation variants can be considered where the application needs localization of symptoms rather than only image-level classification.

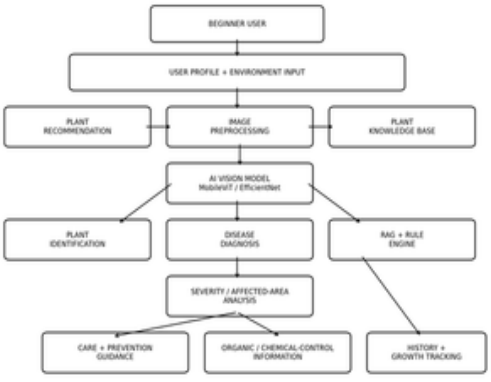
III. PROBLEM STATEMENT AND RESEARCH GAP

The central problem addressed by Agri Guardian is the lack of a single beginner-oriented system that connects plant selection, plant identification, health diagnosis, severity interpretation, care guidance, and continuous monitoring. A user may receive an accurate disease label from a vision model and still not know what the result means or what should be checked next.

The proposed research gap is therefore not the invention of plant disease classification itself. Instead, Agri Guardian integrates multiple components into a personalized decision-support workflow. The intended contribution is an accessible system in which computer vision provides observations, segmentation provides an interpretable affected-area estimate, a curated knowledge base supplies agricultural information, and an AI assistant converts retrieved information into beginner-friendly guidance.

IV. PROPOSED SYSTEM

Agri Guardian is organized into eight functional modules: (1) User and Plant Profile Management, (2) Plant Recommendation, (3) Plant Identification, (4) AI Disease Diagnosis,



A. User and Plant Profile Management

The application maintains the user's gardening profile and the plants being managed. Relevant inputs include experience level, available space, indoor or outdoor placement, approximate light availability, and plant preferences. A plant profile stores identity, care requirements, health observations, scan history, and growth records.

B. Plant Recommendation

The recommendation module matches user conditions against plant requirements. A rule-based or weighted scoring algorithm can calculate compatibility across sunlight, space, maintenance level, climate, and user preference. The output is a ranked set of suitable plants accompanied by short explanations of why each plant fits the user's situation.

C. Plant Identification

The user can capture or upload a plant image. The image is resized and normalized before inference. A lightweight image-classification model can identify the plant category when the trained dataset supports the requested species. The application then retrieves the corresponding plant profile from the knowledge base.

D. AI Disease Diagnosis

For health diagnosis, a trained image classifier receives a plant or leaf image and predicts the most likely supported condition. Candidate architectures include EfficientNet-B0, MobileNetV3, or MobileViT. Model selection should be based on measured validation performance, model size, inference latency, and robustness on real-world images. The application should display a confidence value and avoid presenting the prediction as a guaranteed diagnosis.

E. Disease Severity Analysis

Severity estimation is separated from disease classification. A segmentation model such as U-Net, or an appropriate modern segmentation model, can generate a mask for the visibly affected region. The system can calculate the ratio of affected pixels to the relevant

leaf or plant region and map the result to categories such as mild, moderate, or severe. The severity score is an image-based estimate and should be clearly distinguished from expert field diagnosis.

F. Treatment and Prevention Recommendation

After diagnosis and severity estimation, the recommendation engine retrieves relevant information from a curated agricultural knowledge base. The information is separated into general care, prevention, organic-control information, and chemical-control information. Chemical-control information should be presented only as evidence-based, label-compliant and region-appropriate information rather than as unverified instructions. The system should encourage professional or local agricultural advice when the image confidence is low or the condition is ambiguous.

G. AI Plant Assistant

A retrieval-augmented generation architecture is proposed for the conversational assistant. The user's question is converted into a search query, relevant records are retrieved from the curated plant-care knowledge base, and the language model generates a response grounded in those records. This reduces the risk of unsupported answers compared with a free-form chatbot and allows the same knowledge base to support diagnosis explanations, prevention guidance, and routine care questions.

H. Plant Monitoring and Growth Tracking

The monitoring module maintains a timeline of plant photographs, scan results, care actions, and user notes. Repeated photographs can be compared to identify visible changes over time. The module can show a simple plant-health history and remind the user to perform routine observations.

V. DEVELOPMENT METHODOLOGY

The mobile client is proposed using Android Studio, Kotlin, and Jetpack Compose. Python is used for dataset preparation, model training, evaluation, and experimentation. PyTorch or TensorFlow can be used for deep learning, while OpenCV supports image preprocessing. Model inference can be deployed through a mobile-compatible format such as TensorFlow Lite or ONNX Runtime depending on the selected training framework. A Firebase-based backend can provide authentication, cloud storage, and structured plant-history data. If a Python inference service is required during development, FastAPI can expose model endpoints. For the final

architecture, the vision model can be optimized for local or edge inference where practical, reducing dependence on continuous network connectivity.

TABLE : PROPOSED TECHNOLOGY STACK

Component	Technology	Purpose
Mobile UI	Kotlin + Jetpack Compose	Android application
Computer vision	OpenCV	Preprocessing and image handling
Classifier	MobileViT / EfficientNet-B0	Plant and disease classification
Segmentation	U-Net / suitable segmentation model	Affected-area estimation
ML framework	PyTorch / TensorFlow	Training and evaluation
Knowledge layer	RAG + curated database	Grounded plant-care information

Backend	Firebase / FastAPI	Storage, authentication, APIs
Deployment	TensorFlow Lite / ONNX Runtime	Mobile or edge inference

VI. DATASET AND MODEL TRAINING

The disease-classification component can begin with publicly available plant-image datasets such as PlantVillage. The published PlantVillage study by Mohanty et al. used 54,306 images covering 14 crop species and 26 diseases or healthy classes [1]. However, controlled datasets can differ substantially from photographs captured by users in homes, balconies, gardens, and outdoor environments. Agri Guardian should therefore use augmentation and, where possible, a separate real-world validation set.

Training should include image resizing, normalization, augmentation, train-validation-test splitting, and class-balance analysis. Candidate classifiers should be compared using accuracy, precision, recall, F1-score, confusion matrices, parameter count, and inference latency. For segmentation, intersection over union (IoU) and Dice coefficient can be reported. The final model should be selected from measured results rather than assuming that the most complex architecture is automatically the best.

VII. EVALUATION AND DISCUSSION

The project evaluation should be performed at both model and application levels. At the model level, classification performance should be measured on a held-out test set. Severity estimation should be evaluated against manually annotated affected regions where such annotations are available. At the application level, inference time, model size, usability, and the correctness of retrieved guidance should be evaluated.

A key evaluation condition is real-world image robustness. A model that performs well on clean laboratory images may encounter complex backgrounds, different lighting, occlusion, multiple leaves, camera blur, and unfamiliar varieties in actual use. Therefore, the project should report these limitations and avoid using benchmark accuracy alone as evidence of real-world diagnostic reliability.

This draft intentionally does not report fabricated accuracy, severity, or user-study values. Once the team completes model training and application testing, the measured results should replace this paragraph with a results table containing the actual values and experimental conditions.

VIII. LIMITATIONS AND SAFETY CONSIDERATIONS

Plant appearance can be influenced by multiple factors that are visually similar, including disease, nutrient imbalance, water stress, lighting, pests, and mechanical damage. Consequently, image-based predictions should be treated as decision support rather than definitive professional diagnosis. The system should provide a low-confidence pathway and recommend additional observation or expert assistance when the evidence is insufficient.

The quality of the recommendation module depends on the coverage and reliability of its knowledge base. RAG improves grounding only when the retrieved documents are appropriate and current. Chemical-control information requires particular care because suitability depends on plant, pest or disease, region, regulations, product label, and application context. The prototype should therefore focus on safe, educational and evidence-grounded information instead of unrestricted automated prescriptions.

IX. CONCLUSION AND FUTURE WORK

Agri Guardian proposes a unified AI-assisted plant-care workflow for beginners who want to grow plants but do not have sufficient gardening knowledge. The system combines personalized plant recommendation, computer-vision-based identification and disease diagnosis, severity estimation, knowledge-grounded care recommendations, conversational assistance, and plant-history tracking. The architecture separates visual inference from knowledge retrieval so that the application can provide both machine-generated observations and understandable explanations.

Future work can include a larger real-world image dataset, improved segmentation, on-device model optimization, environmental sensor integration, personalized growth prediction, and a larger verified agricultural knowledge base. A user study with beginner growers can also evaluate whether the system actually improves understanding and confidence in routine plant care.

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